

2048 x 8 Static Random Access Memory

Features

- 100 nsec Maximum Access Time
- Fully Static Operation:
No Clocks or Strokes Required
- Automatic $\overline{CE}$ Power Down
- Identical Cycle and Access Times
- Single +5V Supply ($\pm 10\%$)
- Pin Compatible with 16K ROMs, EPROMs, and EEPROMs
- Totally TTL Compatible:
All Inputs and Outputs
- Common Data Input and Output
- Three-State Output
- JEDEC Approved Pinout

Description

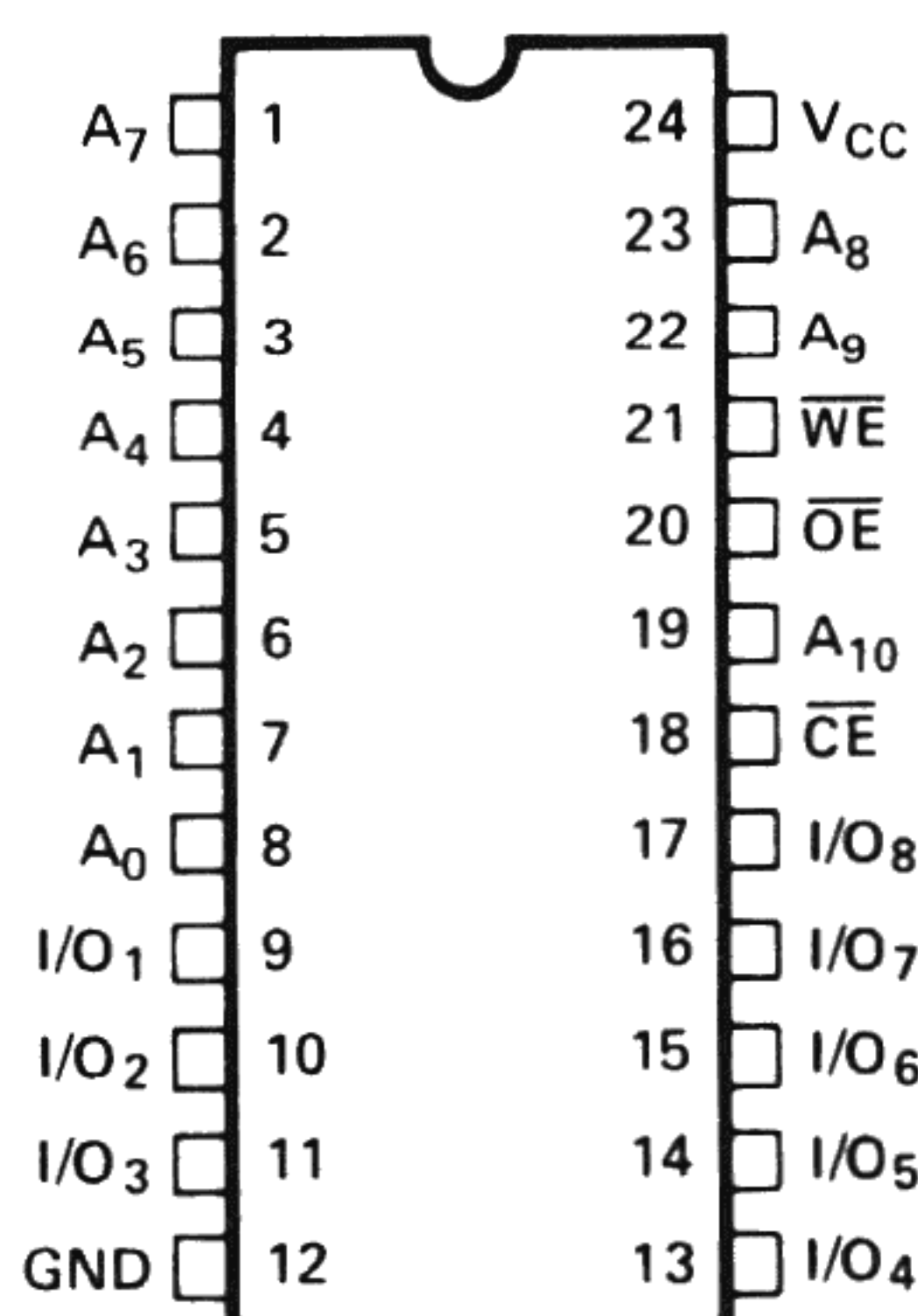
The Synertek SY2128 is a 16,384 bit static Random Access Memory organized 2048 words by eight bits and is fabricated using Synertek's new scaled n-channel silicon gate technology. It is designed using fully static circuitry, therefore requiring no clock or refreshing to operate. The common data input and three-state output pins optimize compatibility with systems utilizing a bidirectional data bus.

The SY2128 offers an automatic power down feature under the control of the chip enable ($\overline{CE}$) input. When $\overline{CE}$ goes high, deselecting the

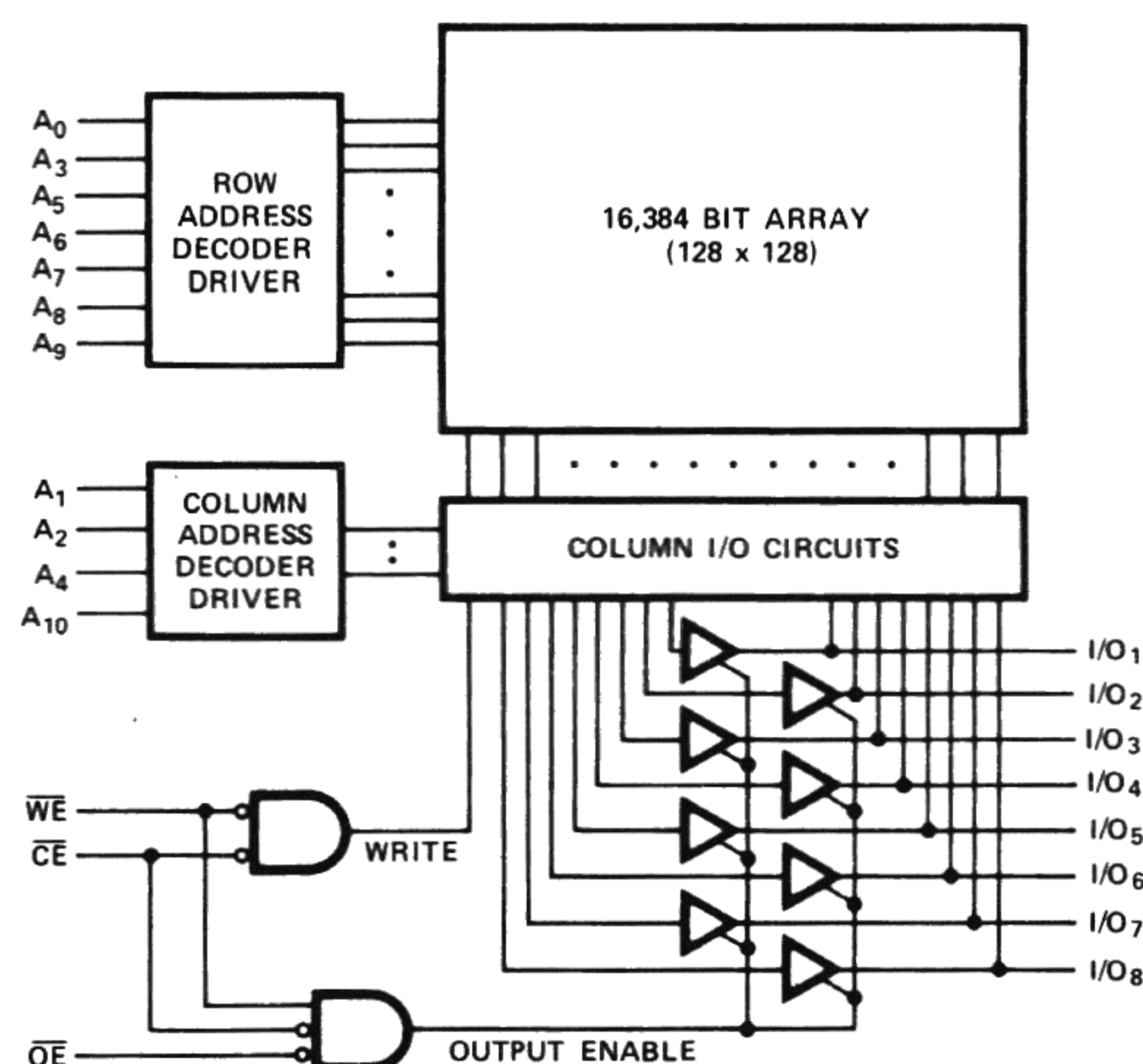
chip, the device will automatically power down and remain in a standby power mode as long as $\overline{CE}$ remains high. This feature provides significant system level power savings.

The SY2128 is configured in the JEDEC approved pinout for 24 pin byte organized memories and is pin compatible with 16K ROMs, EPROMs and EEPROMs. This offers the user the flexibility of being able to switch between RAM, ROM, EPROM, or EEPROM as his needs dictate with a minimum of board changes.

Pin Configuration



Block Diagram



Absolute Maximum Ratings*

Temperature Under Bias -10°C to 85°C
 Storage Temperature -65°C to 150°C
 Voltage on Any Pin with
 Respect to Ground -3.5V to +7V
 Power Dissipation 1.0W
 Electrostatic Discharge Rating (ESD)**
 Inputs to Ground ±2000V

Comment*

Stresses above those listed under "Absolute Maximum Ratings" may cause permanent damage to the device. This is a stress rating only and functional operation of the device at these or any other conditions above those indicated in the operational sections of this specification is not implied.

**Test Condition: MIL-STD-883B Method 3015.1

D.C. Characteristics

$T_A = 0^\circ\text{C to } +70^\circ\text{C}$, $V_{CC} = 5V \pm 10\%$ (Unless otherwise specified)

Symbol	Parameter	2128-1/-2/-3/-4		2128L-1/L-2/L-3/L-4		Unit	Conditions
		Min.	Max.	Min.	Max.		
I_{LI}	Input Load Current (All input pins)		10		10	μA	$V_{CC} = \text{Max}$, $V_{IN} = \text{Gnd to } V_{CC}$
I_{LO}	Output Leakage Current		10		10	μA	$\overline{CE} = V_{IH}$, $V_{CC} = \text{Max}$ $V_{OUT} = \text{Gnd to } 4.5V$
I_{CC}	Power Supply Current		95		75	mA	$T_A = 25^\circ\text{C}$ $V_{CC} = \text{Max}$, $\overline{CE} = V_{IL}$
			100		80	mA	$T_A = 0^\circ\text{C}$ Outputs Open
I_{SB}	Standby Current		20		15	mA	$V_{CC} = \text{Min to Max}$, $\overline{CE} = V_{IH}$
I_{PO}	Peak Power-on Current Note 6		40		30	mA	$V_{CC} = \text{Gnd to } V_{CC} \text{ Min}$ $\overline{CE} = \text{Lower of } V_{CC} \text{ or } V_{IH} \text{ Min.}$
V_{IL}	Input Low Voltage	-3.0	0.8	-3.0	0.8	V	
V_{IH}	Input High Voltage	2.0	6.0	2.0	6.0	V	
V_{OL}	Output Low Voltage		0.4		0.4	V	$I_{OL} = 3.2\text{mA}$
V_{OH}	Output High Voltage	2.4		2.4		V	$I_{OH} = -1.0\text{mA}$

Capacitance

$T_A = 25^\circ\text{C}$, $f = 1.0 \text{ MHz}$

Symbol	Test	Typ.	Max.	Unit
C_{OUT}	Output Capacitance		5	pF
C_{IN}	Input Capacitance		5	pF

NOTE: This parameter is periodically sampled and not 100% tested.

A.C. Characteristics

$T_A = 0^\circ\text{C to } +70^\circ\text{C}$, $V_{CC} = 5V \pm 10\%$ (Note 7)

READ CYCLE

Symbol	Parameter	2128-1/L-1		2128-2/L-2		2128-3/L-3		2128-4/L-4		Unit	Conditions
		Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.		
t_{RC}	Read Cycle Time	100		120		150		200		ns	
t_{AA}	Address Access Time		100		120		150		200	ns	
t_{ACE}	Chip Enable Access Time		100		120		150		200	ns	
t_{AOE}	Output Enable Access Time		35		50		60		70	ns	
t_{OH}	Output Hold from Address Change	10		10		10		10		ns	
t_{LZ}	Output Low Z Time	10		10		10		10		ns	Note 5
t_{HZ}	Output High Z Time	0	35	0	40	0	50	0	60	ns	Note 5
t_{PU}	Chip Enable to Power Up Time	0		0		0		0		ns	
t_{PD}	Chip Disable to Power Down Time		50		60		80		100	ns	

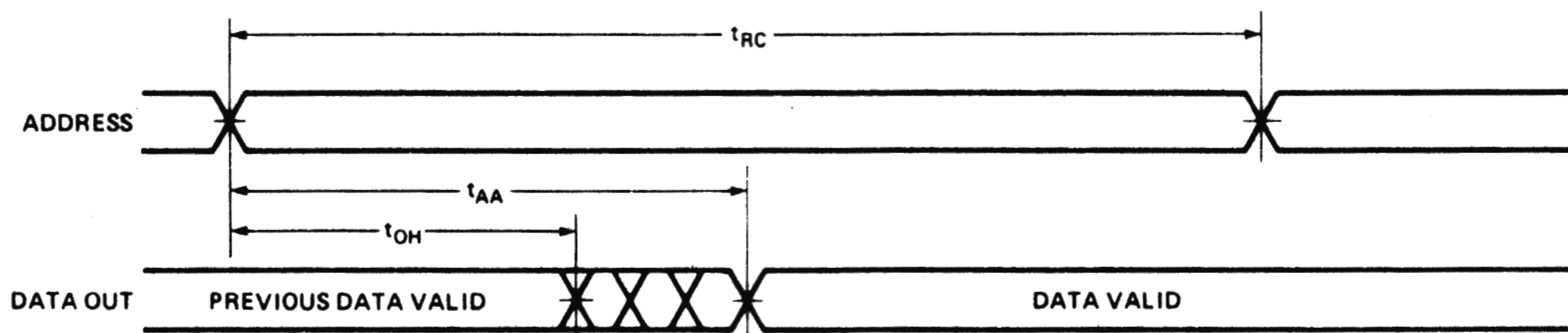
WRITE CYCLE

t_{WC}	Write Cycle Time	100		120		150		200		ns	
t_{CW}	Chip Enable to End of Write	80		90		120		150		ns	
t_{AW}	Address Valid to End of Write	80		90		120		150		ns	
t_{AS}	Address Setup Time	0		0		0		0		ns	
t_{WP}	Write Pulse Width	60		70		90		120		ns	
t_{WR}	Write Recovery Time	0		0		0		0		ns	
t_{DW}	Data Valid to End of Write	40		50		70		90		ns	
t_{DH}	Data Hold Time	0		0		0		0		ns	
t_{WZ}	Write Enabled to Output in High Z	0	35	0	40	0	50	0	60	ns	Note 5
t_{OW}	Output Active from End of Write	0		0		0		0		ns	Note 5

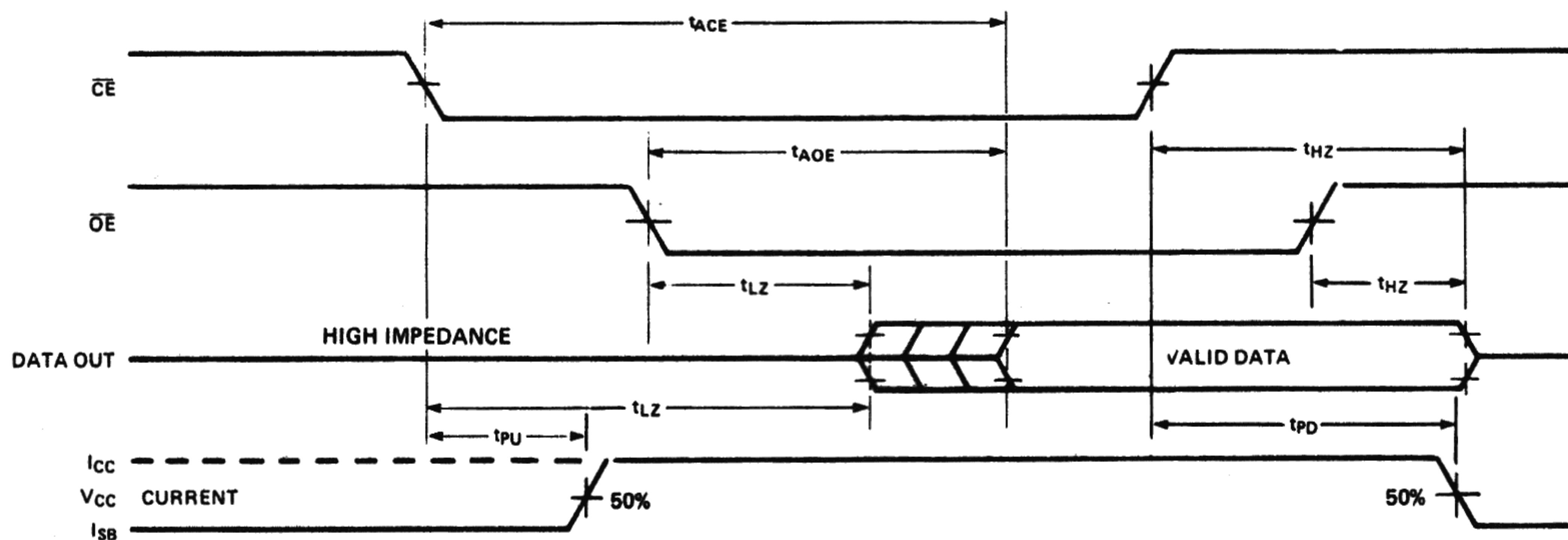
(See following page for notes)

Timing Diagrams

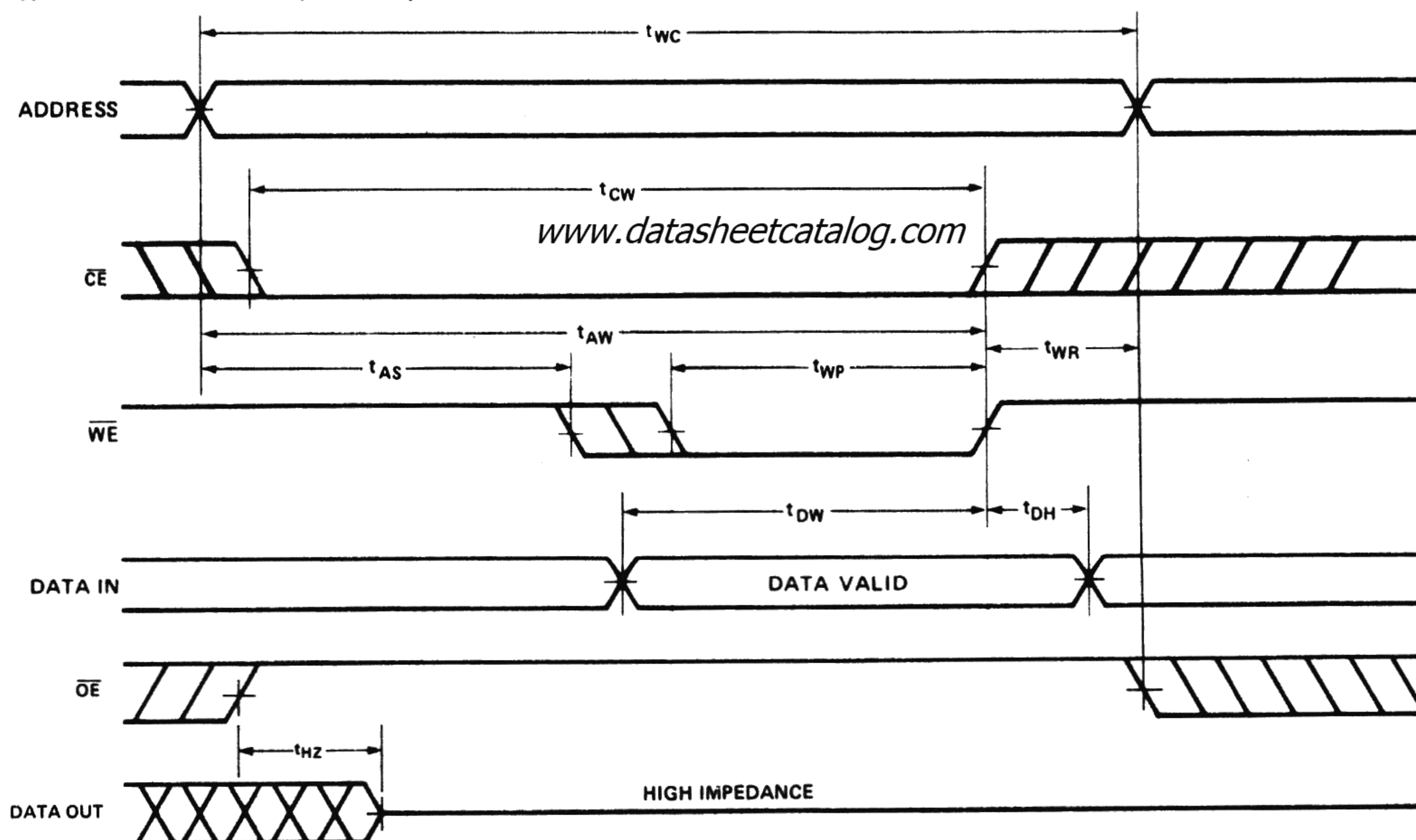
READ CYCLE NO. 1 (NOTES 1 and 2)



READ CYCLE NO. 2 (NOTES 1 and 3)

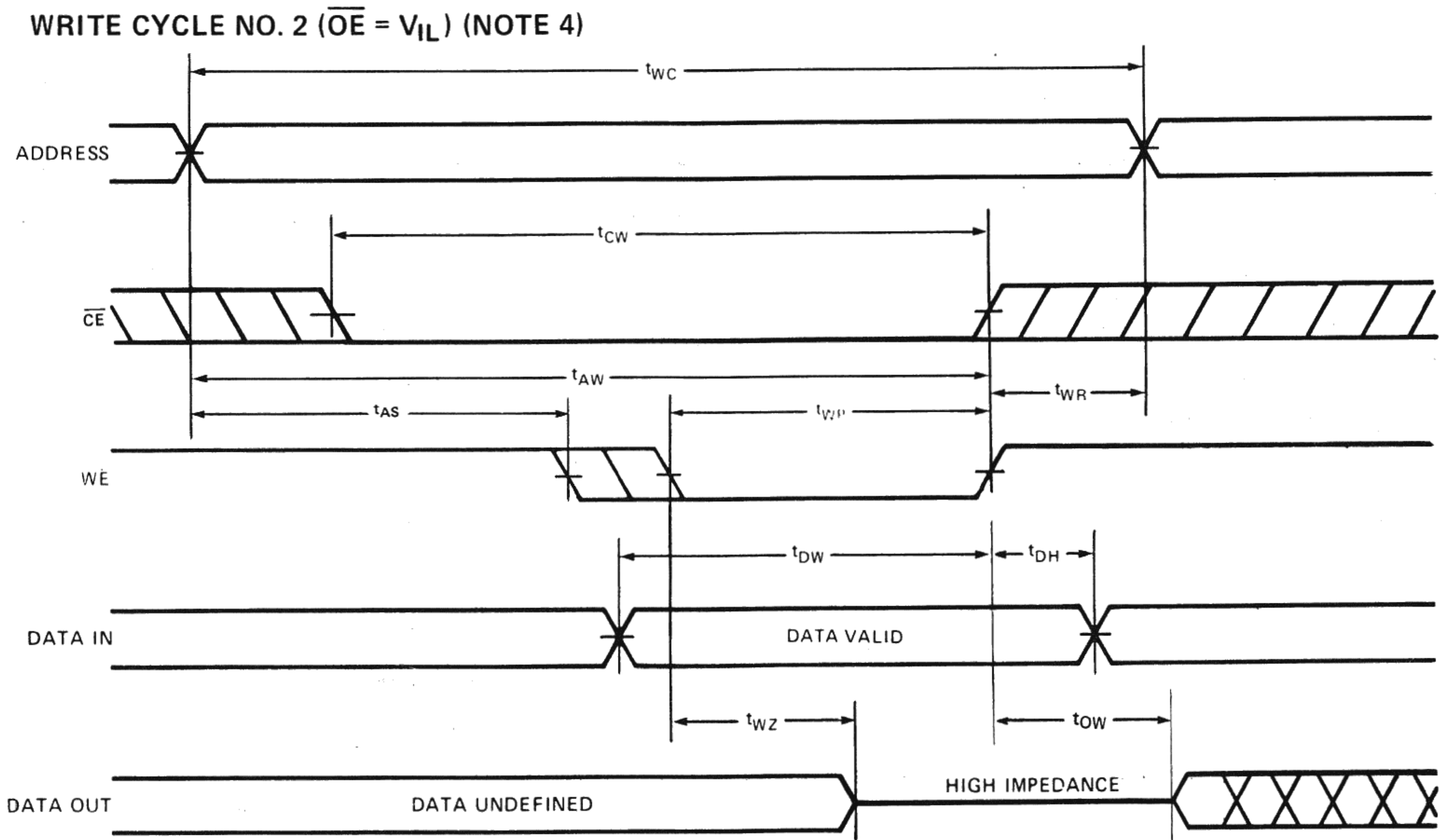


WRITE CYCLE NO. 1 (NOTE 4)

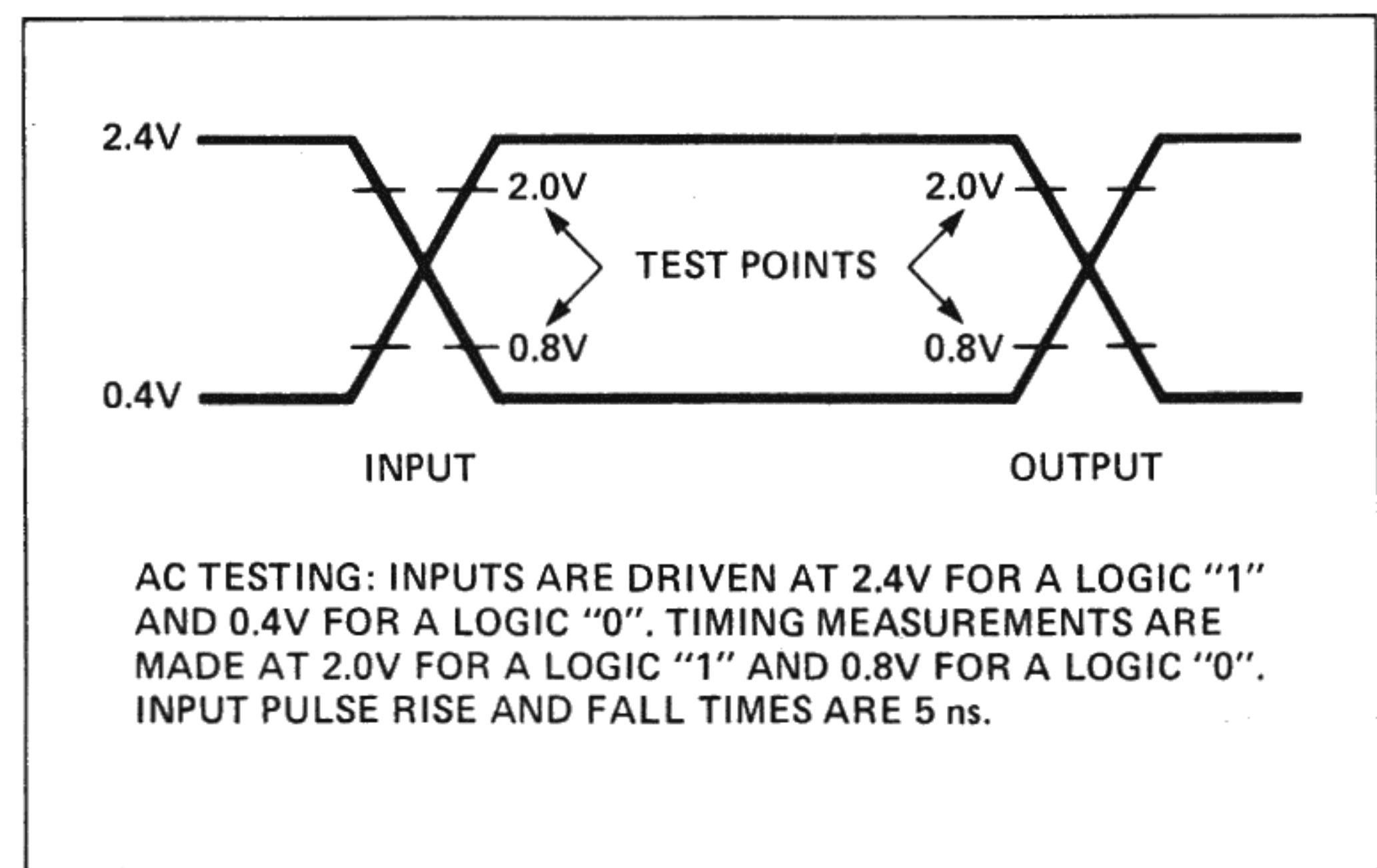


Notes:

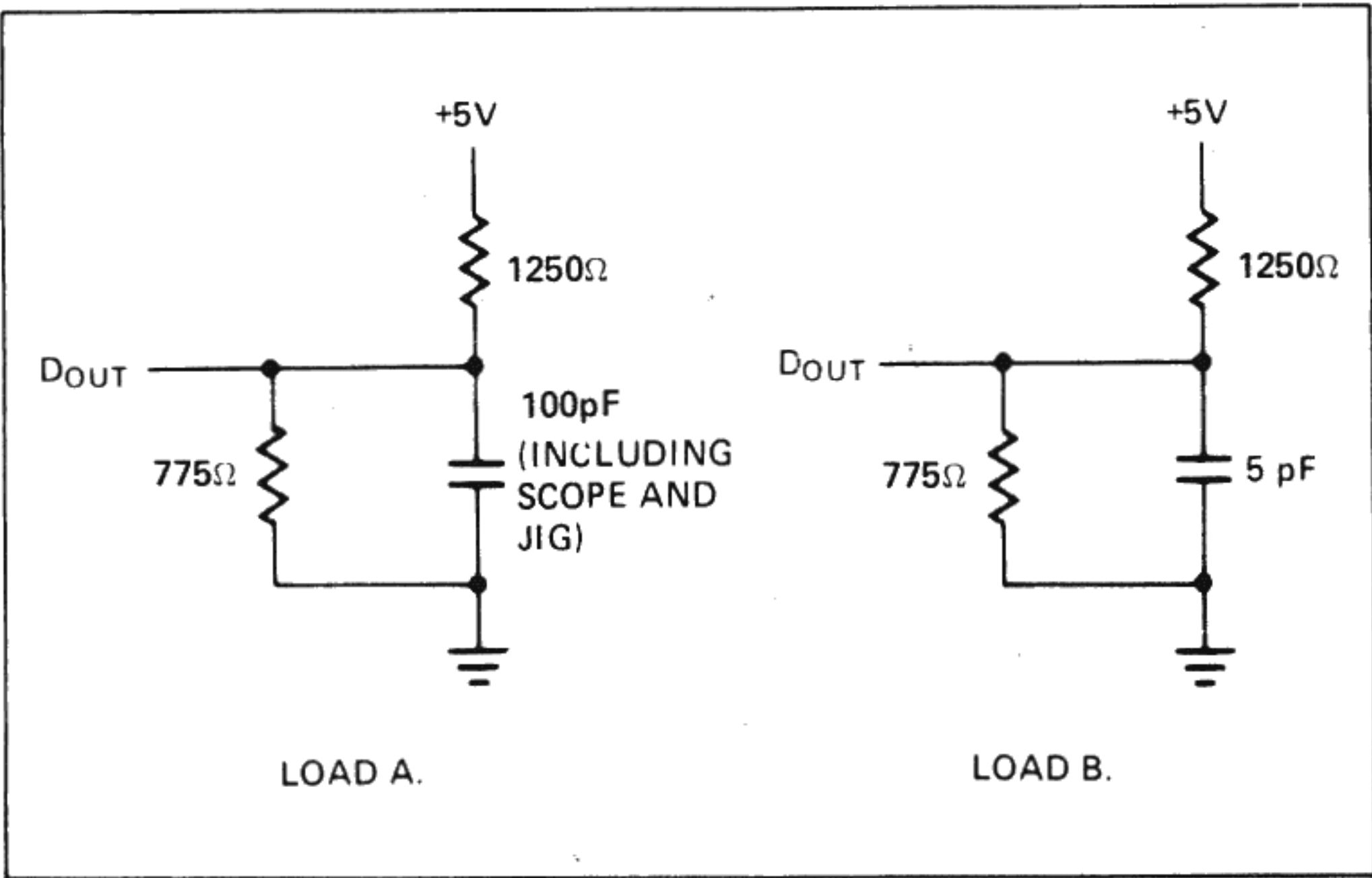
1. $\overline{WE}$ is high for Read Cycles.
2. Device is continuously selected, $\overline{CE} = \overline{OE} = V_{IL}$.
3. Addresses valid prior to or coincident with $\overline{CE}$ transition low.
4. If $\overline{CE}$ goes high simultaneously with $\overline{WE}$ high, the outputs remain in the high impedance state.
5. Transition is measured $\pm 500\text{mV}$ from low or high impedance voltage with load B. This parameter is sampled and not 100% tested.
6. A pullup resistor to V_{CC} on the $\overline{CE}$ input is required to keep the device deselected: otherwise, power-on current approaches I_{CC} active.
7. A minimum 0.5 ms time delay is required after application of V_{CC} (+5V) before proper operation is achieved.



A.C. Testing Input, Output Waveform



A.C. Testing Load Circuit



Package Availability 24 Pin Cerdip
24 Pin Plastic

Ordering Information

Order Number	Access Time (Max)	Operating Current (Max)	Standby Current (Max)	Package Type
SYD2128-1	100ns	100mA	20mA	Cerdip
SYP2128-1	100ns	100mA	20mA	Plastic
SYD2128-2	120ns	100mA	20mA	Cerdip
SYP2128-2	120ns	100mA	20mA	Plastic
SYD2128-3	150ns	100mA	20mA	Cerdip
SYP2128-3	150ns	100mA	20mA	Plastic
SYD2128-4	200ns	100mA	20mA	Cerdip
SYP2128-4	200ns	100mA	20mA	Plastic
SYD2128L-1	100ns	80mA	15mA	Cerdip
SYP2128L-1	100ns	80mA	15mA	Plastic
SYD2128L-2	120ns	80mA	15mA	Cerdip
SYP2128L-2	120ns	80mA	15mA	Plastic
SYD2128L-3	150ns	80mA	15mA	Cerdip
SYP2128L-3	150ns	80mA	15mA	Plastic
SYD2128L-4	200ns	80mA	15mA	Cerdip
SYP2128L-4	200ns	80mA	15mA	Plastic